

Evaluation of Tumor Regression and Other Prognostic Factors for Early and Late Metastasis after Proton Irradiation of Uveal Melanoma

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Abstract: The authors examined the relationship of change in tumor height after proton beam irradiation with the risk of metastasis in 700 patients treated for uveal melanoma before July 1, 1986. Rates of change in tumor height were computed for each patient using follow-up ultrasonographic measures during the first 2 years after treatment. Risk of metastasis was evaluated separately in the first 2 years after treatment when tumor decline was assessed (concurrent metastasis), and 2 years or later after treatment (subsequent metastasis). Using Cox proportional hazards models to adjust for other known risk factors, tumors regressing rapidly were significantly more likely to metastasize concurrently with their regression ($P < 0.05$). The opposite was found for subsequent metastasis: tumors with a slow annual decline were more likely to metastasize after 2 years after treatment ($P < 0.05$). Substantial changes also were noted in the ability of previously described prognostic factors (largest tumor diameter, tumor height, ciliary body involvement, and patient's age) to predict early and late metastasis. Characteristics placing individuals at high risk of metastasis may change markedly with time after treatment. *Ophthalmology* 96:1566-1573, 1989

It has been proposed that rapid decline in tumor height after irradiation of a uveal melanoma may be associated with a greater risk of subsequent metastasis.¹ Supporting this hypothesis are reports that faster growing tumors be-

fore treatment regress more rapidly than slower growing tumors² and mixed cell posterior uveal melanomas, known to have poorer prognosis than spindle tumors, regress more rapidly than spindle cell melanomas.³ A contradictory report found faster tumor regression in spindle cell tumors, as assessed by acoustic tissue type, compared with mixed/epithelioid cell type.⁴ One study found shorter tumor height at 12 months, adjusting for pretreatment height, to be associated with a higher risk of death due to metastasis.¹ Follow-up measures of tumor height at different times after treatment and measures of percentage change in tumor height were not significant predictors controlling for pretreatment characteristics.¹

We have recently described measures of absolute and

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